E.C. - Math challenge #1
ENGR& 204 #17385

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1 Problem 1.

A trapezoid has consecutive sides of length 8, 10, and 12 in that order. Find the largest possible area of such a trapezoid rounded down to the nearest integer.

DEFINITION 1: AREA OF A TRAPEZOID

$$A = \frac{a+b}{2}h \tag{1}$$

SOLUTION: When approaching an optimization problem, we are going to create an area function and evaluate the vertices. In this geometry problem, it important to realize that there are two possible variations on how we can create a trapezoid.

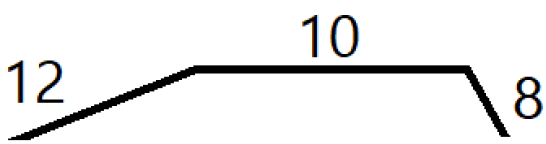


Figure 1: Slant-Base-Slant Trapezoid

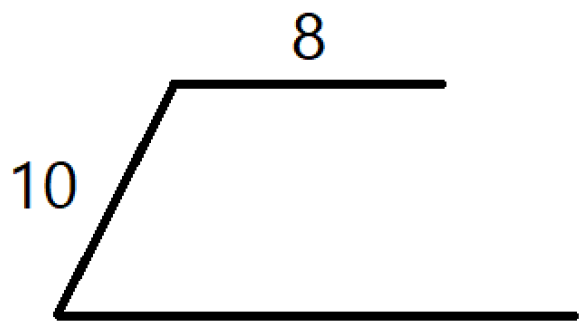


Figure 2: Base-Slant-Base Trapezoid

We can either create a Slant-Base-Slant or a Base-Slant-Base trapezoid. Our first problem is to decide whether each case gives out a larger area. Let us first start with the Base-Slant-Base case. Since we have fixed bases, we can simply use our definition for area of a trapezoid. To maximize the area also implies maxing out the height if and only if the bases are fixed. More formally, we can describe it with mathematical terms.

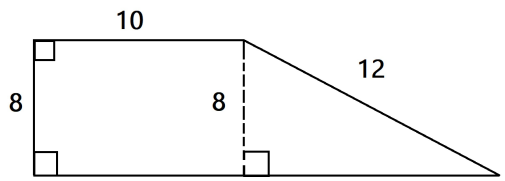
For a fixed $a, b \in \mathbb{R}^+$, it implies the following:

$$A(h) = \frac{a+b}{2}h$$
$$\implies \arg\max_h A = \arg\max_h h$$

From this, we can logically find the maximum area for our Base-Slant-Base trapezoid by making it a right trapezoid.

$$
\begin{aligned}
a = 8,\ b = 12,\ h = 10 \\
A &= \frac{a+b}{2}h \\
A_{bsb\ maxed} &= \frac{8+12}{2}10 \\
\implies A_{bsb\ maxed} &= 100 \text{ units}^2
\end{aligned}
$$

Our next case is the Slant-Base-Slant trapezoid. Something to note is that if we were to maximize height, we can show that the area of a Slant-Base-Slant trapezoid is larger than the area of a Base-Slant-Base trapezoid. This means that our answer is actually optimizing the Slant-Base-Slant triangle.



$$
\begin{aligned}
s_1 = 8,\ a = 10, s_2 = 12 \\
A_{sbs} &= 8 \cdot 10 + \frac{8 \cdot (\sqrt{12^2 - 8^2})}{2} \\
&= 80 + 4\sqrt{16 \cdot 5} \\
&= 80 + 16\sqrt{5} \\
A_{sbs} &\approx 115.777 \text{ units}^2 \\
\implies &\boxed{A_{sbs} > A_{bsb\ maxed}}
\end{aligned}
$$

1.1 Optimizing Slant-Base-Slant Trapezoid

Just as a sanity check, it is imperative that we define the bounding range of the opposite base. If we were to collapse the two slants into a line, we would approach a max base would be 30. With the opposing argument, if we were to connect the two slants into a point, our minimum base would be 0. Using Heron's Formula, we have a "minimum" area of $A = \sqrt{s(s-a)(s-b)(s-c)} = \sqrt{15(15-8)(15-10)(15-12)} = 15\sqrt{7} \approx 39.7$ units^2. All of this information should give us a base to work with.

In hindsight, I'd like to work with area function that is dependent on height. We know off the bat that it is restricted from $0 < x \leq 8$ from the triangle approach and from maximizing one of the slant leg height. We can solve for the other base in terms of the height using the Pythagorean Theorem and some substitutions. After which, we can create our area function and maximize it.

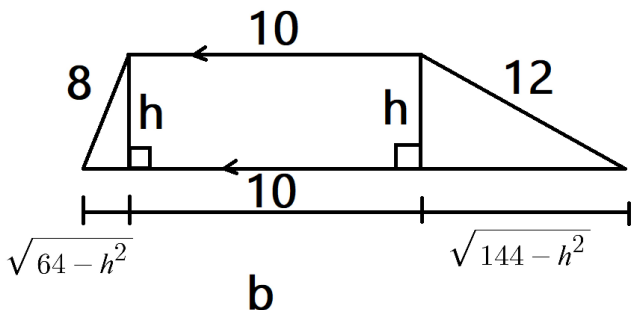


Figure 3: Slant-Base-Slant Trapezoid

$$A = \frac{a+b}{2}h$$

$$A = \frac{10+b}{2}h, \quad b = \sqrt{64-h^2} + 10 + \sqrt{144-h^2}$$

$$A(h) = \frac{10 + \left(\sqrt{64-h^2} + 10 + \sqrt{144-h^2}\right)}{2}h, \quad 0 < h \leq 8$$

$$A'(h) = 10 + \frac{\sqrt{64-h^2}}{2} - \frac{h^2}{2\sqrt{64-h^2}} + \frac{\sqrt{144-h^2}}{2} - \frac{h^2}{2\sqrt{144-h^2}}$$

$$A'(h) = 0 \Longrightarrow h_{max\ area} \approx 7.6552$$

$$A(h_{max\ area}) \approx 120.816$$

$$\Longrightarrow \boxed{\lfloor A(h_{max\ area}) \rfloor = 120}$$

We are kind of forced to use numerical methods as solving for when deriviative is equal to 0 would be quite the algebraic hassle. Another way would be to floor the entire $A(h)$ function and simply graph values.

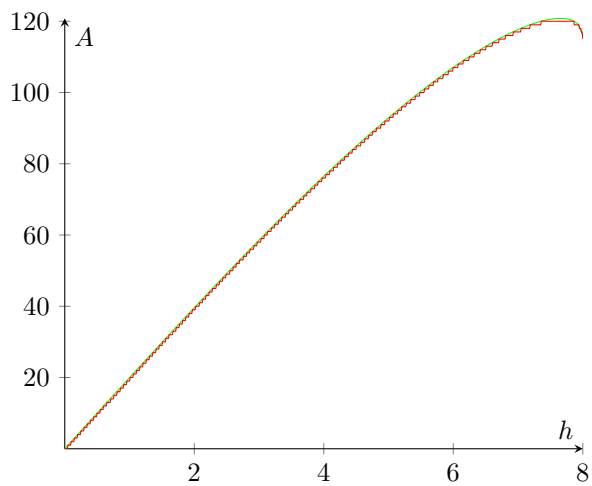


Figure 4: Graph of both normal and floor functions.
$A(h) = \frac{10+\left(\sqrt{64-h^2}+10+\sqrt{144-h^2}\right)}{2}h, \quad 0 < h \leq 8$

h	$\lfloor A(h) \rfloor$
7.0	117
7.1	118
7.2	119
7.3	119
7.4	120
7.5	120
7.6	120
7.7	120
7.8	120
7.9	119
8.0	115

Figure 5: h vs $\lfloor A(h) \rfloor$ from [7, 8]

To conclude, a trapezoid with consecutive sides of length 8, 10, and 12 in that order has the largest possible area rounded down to the nearest integer of $\boxed{120 \text{ units}^2}$.

2 Problem 2.

Each member of a group of 6 people is either a knight (who always tells the truth), or a knave (who always lies). The first person says, "There is at least one knave in this group," and the nth person ($2 \leq n \leq 6$) says that there are at least n knaves in the group. How many knaves are there?

SOLUTION: These logic problems usually emerges a pattern with higher iterations. I like to start with smaller cases and to see we can build up to induction. Let us start with the case of $n = 1$. We see in this odd case, it is a paradox and there is no answer. Let us move on to $n = 2$. We start at the person who says the largest n is a knave. Logically, if n are a knave, that would also have to include itself which would be the truth which would be a contradiction. Therefore, the n person is a knave. This would make the first person 1 to become a knight. So an even split.

- P1: at least 1 knave $\longrightarrow$ knight
- P2: at least 2 knaves $\longrightarrow$ knave

If we were to up the n, we would see this pattern repeat. In whole, the pattern is to work from the largest n which results in a knave, then go to the smallest 1 which is a knight. Then, loop back to largest which is $n-1$ which results in a knave and the smallest 2 which is a knight. We continually loop back and this either results in a paradox (n is odd) or an even split of knights and knaves at the halfway point (n is even). This brings up a good point to the paradoxical odd n case as we can say that the $\lceil \frac{n}{2} \rceil$ person is both half knave and half knight.

Here is our solution then.

$$Knaves(n) = \begin{cases} DNE, & n \equiv 1 \pmod 2 \\ \frac{n}{2}, & n \equiv 0 \pmod 2 \end{cases}, \ n \in \mathbb{N}$$

$$\Longrightarrow \boxed{Knaves(6) = 3}$$

3 Problem 3.

The two distinct lines with equations $ax + y = c$ and $x + y = c$ intersect at the point $(b - 2, 4)$. Solve for $b + c$.

SOLUTION: Okay, this problem seems much easier than the previous ones. From the wording alone, it seems to hint that a should be arbitriary with the sole exception of the $a \neq 1$ to satisfy the distinct line requirement. Let us represent the system of equations with matrices and into a linear combination of vectors.

$$\begin{aligned} ax + y &= c \\ x + y &= c \\ \implies \begin{bmatrix} a & 1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} &= \begin{bmatrix} c \\ c \end{bmatrix} \\ \begin{bmatrix} a \\ 1 \end{bmatrix} x + \begin{bmatrix} 1 \\ 1 \end{bmatrix} y &= \begin{bmatrix} c \\ c \end{bmatrix} \end{aligned}$$

Since the vectors are both linearly independent or in other words the null space is the zero vector, we are guarenteed only one solution when $x = 0,\ y = c$. We can show this with Column-Row factorization of the augmented matrix.

$$\begin{bmatrix} a & 1 & c \\ 1 & 1 & c \end{bmatrix} = \begin{bmatrix} a & 1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & c \end{bmatrix}$$

The factorization shows us also the row reduced echelon form which gets us an intersection point at $(0, c)$. We simply just need to match this with our given intersection point with $(b - 2, 4)$.

$$\begin{aligned} &(0, c) = (b - 2, 4) \\ &0 = b - 2 \\ &b = 2 \\ &c = 4 \\ \implies &\boxed{b + c = 6} \end{aligned}$$